MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG4SECECT200 Circuit Simulation and PCB Design

Duration: 1.5hours **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability.*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries 1 mark

1	What is the main purpose of a Printed Circuit Board (PCB)? A. To generate electrical power B. To mechanically support and electrically connect components C. To store electrical energy D. To amplify signals	U	1
2	Which type of PCB has conductive tracks only on one side of the board? A. Multilayer PCB B. Double-sided PCB C. Single-sided PCB D. Flexible PCB	U	1
3	A double-sided PCB differs from a single-sided PCB because it: A. Uses plastic substrate B. Has components only on one side C. Has copper layers on both sides D. Cannot support ICs	U	1
4	Multilayer PCBs are mainly used to: A. Reduce component cost B. Increase board thickness C. Support complex circuits with higher component density D. Eliminate soldering	U	1
5	In PCB component identification, the prefix “R” generally represents: A. Capacitor B. Resistor C. Diode D. Inductor	U	1

6	Which symbol is commonly used to label capacitors on a PCB? A. R B. L C. C D. D	U	1
7	Surface Mount Devices (SMDs) are preferred because they: A. Require holes for mounting B. Are larger in size C. Allow high component density and compact design D. Cannot be automated	U	1
8	In SMD resistor identification, the value is usually indicated by: A. Colour bands B. Printed alphanumeric codes C. Size of the component D. Number of terminals	U	1
9	The most commonly used PCB substrate material is: A. Aluminium B. Copper C. FR-4 (Glass epoxy) D. Silicon	U	1
10	Copper cladding in a PCB is used to: A. Insulate component B. Increase board flexibility C. Form conductive tracks D. Protect from heat	U	1
11	Which file format is universally used for PCB manufacturing output? A. PDF B. DOCX C. GERBER D. JPEG	U	2
12	The main purpose of design rules in PCB layout is to: A. Improve colour appearance B. Reduce software size C. Ensure manufacturability and reliability D. Increase board thickness	U	2
13	Raw PCB cutting is performed mainly to: A. Improve conductivity B. Obtain the required board dimensions C. Remove copper layers D. Reduce thermal losses	U	2
14	Surface cleaning of a PCB before printing is necessary to: A. Increase drilling speed B. Remove oxidation, grease, and dust C. Increase copper thickness	U	2

	D. Improve enclosure fitting		
15	Marking of mounting screw positions on a PCB is done to: A. Improve circuit speed B. Reduce heat generation C. Ensure proper mechanical fixing D. Avoid short circuits	U	2
16	Thermal print transfer in PCB fabrication is used for: A. Drilling holes B. Transferring circuit layout onto copper surface C. Applying protective coating D. Electrical testing	U	2
17	Copper etching is the process of: A. Depositing copper on substrate B. Removing unwanted copper from the board C. Drilling component holes D. Applying solder mask	U	2
18	Through-hole plating is mainly required to: A. Increase board thickness B. Electrically connect copper layers C. Improve solder mask quality D. Reduce drilling time	U	2
19	Automated drilling machines are preferred in PCB manufacturing because they: A. Require manual alignment B. Are slower than hand drilling C. Provide high accuracy and repeatability D. Cannot drill small holes	U	2
20	Continuity, short, and open circuit tests are performed to: A. Improve visual appearance B. Increase copper adhesion C. Verify electrical correctness of PCB D. Select enclosure type	U	2
21	Which of the following is a commonly used PCB design and simulation software? A. MS Word B. EasyEDA C. Photoshop D. AutoCAD Civil	U	3
22	Proteus software is mainly used for: A. Mechanical drafting B. Circuit simulation and PCB design C. Text editing D. Image processing	U	3
23	Schematic capturing refers to: A. Physical assembly of components B. Drawing the electrical circuit diagram	U	3

	C. Drilling PCB holes		
	D. Applying solder mask		
24	Ohm’s Law is mathematically expressed as:	U	3
	A. P = VI		
	B. V = IR		
	C. R = VI		
	D. I = VR		
25	A voltage divider circuit is used to:	U	3
	A. Increase current		
	B. Divide input voltage into smaller voltages		
	C. Amplify signals		
	D. Reduce resistance		
26	In an LED circuit, a resistor is connected in series to:	U	3
	A. Increase brightness indefinitely		
	B. Protect LED from excessive current		
	C. Change LED colour		
	D. Increase voltage		
27	Selection of components in simulation software mainly depends on:	U	3
	A. Colour of components		
	B. Electrical specifications and package type		
	C. Software theme		
	D. Board size only		
28	A footprint in PCB design represents:	U	3
	A. Electrical symbol of component		
	B. Physical layout of component pads		
	C. Power rating of component		
	D. Simulation waveform		
29	Assigning footprints to components is required to:	U	3
	A. Simulate voltage levels		
	B. Enable PCB layout design		
	C. Increase circuit speed		
	D. Reduce file size		
30	Net listing in PCB design is the process of:	U	3
	A. Listing component prices		
	B. Defining electrical connections between pins		
	C. Generating silkscreen text		
	D. Exporting PDF files		
31	Auto routing in PCB design means:	U	3
	A. Manual drawing of tracks		
	B. Software automatically routes tracks		
	C. Drilling holes automatically		
	D. Automatic soldering		
32	Manual routing is preferred when:	U	3
	A. Board is very simple		

	B. High-speed or sensitive signals are present		
	C. Software is unavailable		
	D. BOM generation is required		
33	Silkscreen layer in a PCB is used to display:	U	3
	A. Copper tracks		
	B. Component labels and markings		
	C. Solder joints		
	D. Drill holes		
34	Bill of Materials (BOM) contains details of:	U	3
	A. Copper thickness		
	B. Drill sizes		
	C. Components used in the PCB		
	D. Gerber layers		
35	Soldering in PCB assembly is the process of:	U	3
	A. Removing components		
	B. Electrically and mechanically joining components		
	C. Designing circuit diagrams		
	D. Testing PCB continuity		
		1 * 35 = 35	

Part B

Short Answer Questions

Answer any **3** Questions

Each question carries 5 marks

36	Discuss the types of PCBs	U	1
37	Describe the identification and labelling of PCB components.	U	1
38	Explain the basic design rules for PCB design	U	2
39	Describe the PCB fabrication process	U	2
40	Explain the PCB layout designing using simulation software.	U	3
		5 * 3 = 15	